# Hotel Revenue Analysis (2018–2020)

This comprehensive data analytics project examines hotel booking patterns, revenue trends, and operational performance across a three-year period from 2018 through 2020. The analysis leverages booking data to identify revenue opportunities and operational inefficiencies.


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# Problem Statement

## Project Context

The hospitality industry faces increasing pressure to optimize revenue while maintaining guest satisfaction. Hotel operators need to understand booking patterns, seasonal trends, and pricing strategies to maximize profitability. Without comprehensive data analysis, revenue management decisions remain reactive rather than proactive.

## Core Challenges

- Revenue management decisions based on limited historical data
- Difficulty identifying high-value customer segments
- Challenges in optimizing room pricing strategies
- Limited visibility into booking lead times and cancellation patterns
- Need for data-driven insights to guide capacity planning

# Project Objectives: A Strategic Roadmap

This analysis was designed to address five critical business questions that would guide strategic decision-making for hotel management. Each objective was carefully crafted to extract maximum value from the available data.

### Analyze Revenue Trends

Examine temporal patterns in revenue generation across different time periods, seasons, and market conditions to identify growth opportunities and risk factors.

### Study Booking Behavior

Understand how customers book rooms, their lead times, booking channels, and cancellation patterns to optimize reservation strategies.

### Understand Customer Segmentation

Identify distinct customer groups based on demographics, booking patterns, and spending behavior to enable targeted marketing approaches.

### Evaluate Pricing Strategies

Analyze average daily rates, revenue per available room, and pricing elasticity across different segments and time periods.

### Generate Actionable Insights

Transform data findings into concrete recommendations that hotel management can implement immediately to improve performance metrics.

# Dataset Overview: The Foundation of Analysis

## Comprehensive Data Coverage

- The dataset spans three years (2018-2020), capturing a complete business cycle that includes seasonal variations and year-over-year performance.
- This temporal breadth enables trend analysis and year-to-year comparisons.
- The data encompasses both city hotels and resort properties, providing insights into different market segments and operational models.
- Key data elements include booking details, revenue figures, average daily rates, occupancy rates, market segments, customer types, booking channels, and stay durations.
- This comprehensive coverage allows for multi-dimensional analysis across various business dimensions.

**3**

**Years of Data**

2018-2020 coverage

**86.8K**

**Total Bookings**

Complete reservation records

**$30.5M**

**Revenue Generated**

Accumulated earnings

**3.5**

**Average Stay**

Days per booking

# Data Preprocessing

Raw booking data requires extensive preparation before analytical analysis can commence. The preprocessing phase ensures data quality, consistency, and analytical readiness through systematic cleaning and transformation procedures.

01

## Data Cleaning

Remove irrelevant fields, correct inconsistent naming conventions, and standardize data formats across all attributes to ensure uniformity.

02

## Missing Value Handling

Identify and address missing data points through appropriate imputation methods or exclusion criteria based on data importance and pattern analysis.

03

## Duplicate Removal

Detect and eliminate duplicate booking records that may skew analytical results and create artificial inflation in metrics.

04

## Data Transformation

Create derived attributes such as booking lead time, revenue per booking, and customer tenure to enhance analytical capabilities.

05

## Calculated Measures

Develop business logic calculations using DAX including occupancy rates, revenue per available room, and customer lifetime value metrics.

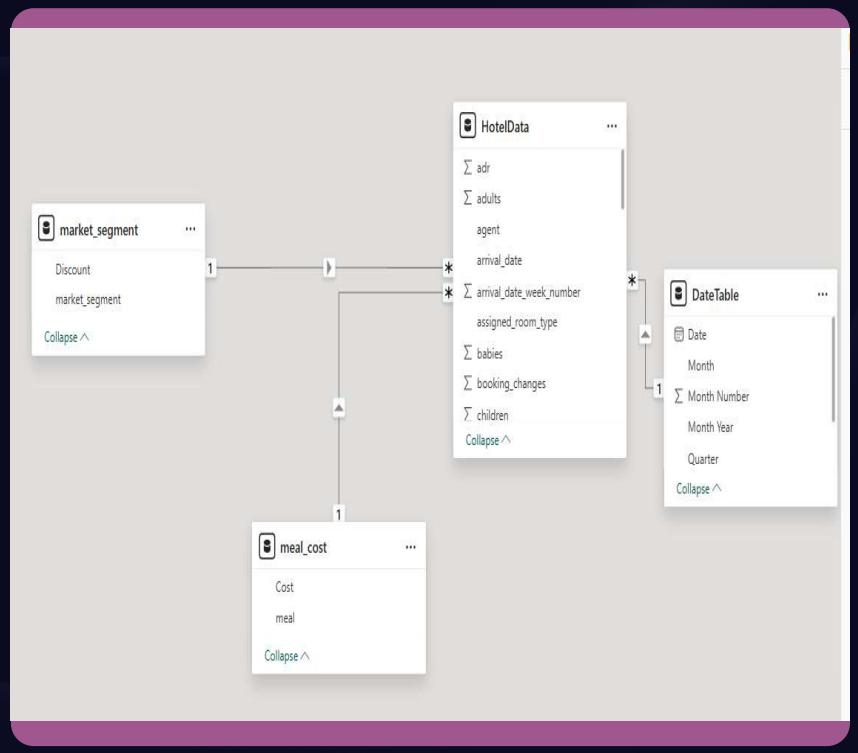
# Data Modeling

## Model Architecture

The data model establishes relationships between booking facts and dimension tables including customer demographics, room types, and temporal attributes. Calculated measures enable revenue analysis across multiple dimensions.

## Key Components

- Fact table storing booking transactions
- Dimension tables for customer attributes
- Temporal dimensions for time-based analysis
- Room type and property classifications
- Booking channel and source tracking



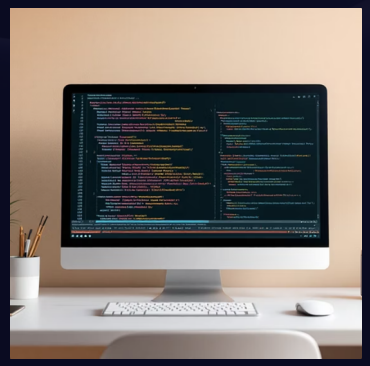
# Tools and Technologies

## Power BI

Primary platform for dashboard creation, interactive visualization, and business intelligence reporting. Enables drag-and-drop analytics and enterprise-grade data connectivity.

## Microsoft Excel

Initial data handling, exploratory analysis, and data preparation tasks. Provides flexible environment for data manipulation and preliminary statistical analysis.



## DAX (Data Analysis Expressions)

Calculation language for creating custom measures, KPIs, and business logic. Enables complex analytical computations and time-intelligence calculations.

# Dashboard Architecture

The analytical solution comprises four interconnected dashboards, each designed to address specific business questions while maintaining consistent visual design and data integration. This multi-dashboard approach enables users to drill from high-level executive summaries to detailed operational insights seamlessly.

### Executive Overview

High-level revenue metrics, booking performance, and occupancy trends for C-level decision-makers. Includes year-over-year comparisons and key performance indicators.

### Booking Behavior Analysis

Detailed examination of customer booking patterns, channel performance, lead time analysis, and cancellation rates. Reveals customer decision-making processes.

### Revenue & Pricing Analysis

ADR trends, revenue per available room calculations, pricing strategy effectiveness, and market segment profitability comparisons.

### Operational Insights

Stay duration patterns, seasonal demand forecasting, resource utilization metrics, and operational efficiency benchmarks.

Each dashboard serves distinct stakeholder groups while sharing common data sources and calculated measures. This architecture balances analytical depth with usability, allowing users to explore data at appropriate levels of detail for their roles.

# Executive Overview Analysis

| 30.5M | 86.8K | $100.8 | 28.9% | 3.5 |
|---|---|---|---|---|
| Total Revenue | Total Bookings | Average Daily Rate | Occupancy Rate | Average Stay |
| Generated 2018-2020 | Complete reservations | Per room night | Rooms filled | Days per booking |

## Key Insights

- **Strong revenue generation** demonstrates effective market positioning and demand capture during the analyzed period. The $30.5M revenue reflects consistent booking performance across both hotel types.
- **Low occupancy rate (28.9%)** indicates significant unused capacity and potential revenue loss. This suggests opportunities for improved yield management and demand stimulation strategies.
- **Seasonal trends visible** in monthly revenue patterns show predictable peaks during summer months and holiday seasons, enabling better resource planning and dynamic pricing implementation.

Conclusion: This indicates that while revenue performance is strong, improving occupancy and leveraging seasonal demand patterns can further enhance overall profitability.

# Booking Behavior Analysis

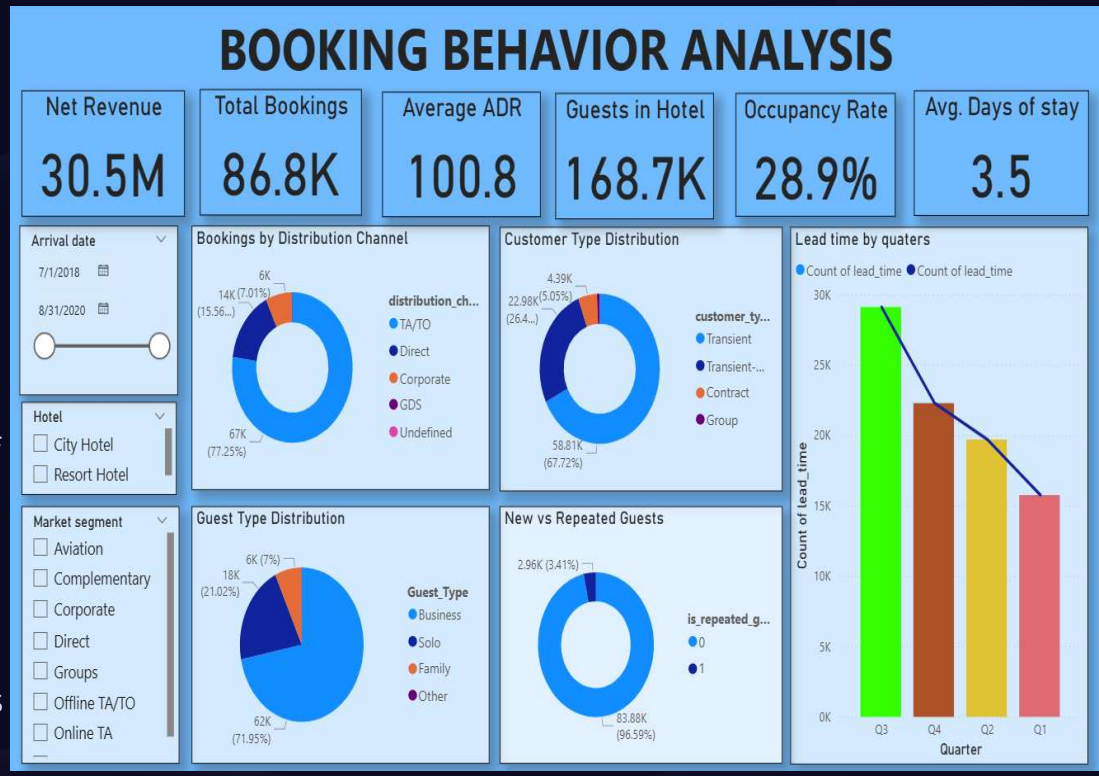
This dashboard focuses on analyzing customer booking behavior and reservation patterns. It helps in understanding how different types of customers interact with hotel booking channels and services.

## Key Metrics

- Transient customers dominate bookings (~67%)
- New customers account for ~96% of total bookings, while repeat customers are very low (~3%)
- Travel agents (TA/TO) contribute approximately ~77% of total bookings

## Strategic Insights

- The hotel is highly dependent on new customers
- Customer retention is significantly low
- There is strong reliance on third-party booking platforms




**Conclusion:** This indicates that improving customer retention and increasing direct bookings are essential for long-term profitability.

# Revenue & Pricing Analysis

## Analysis Overview

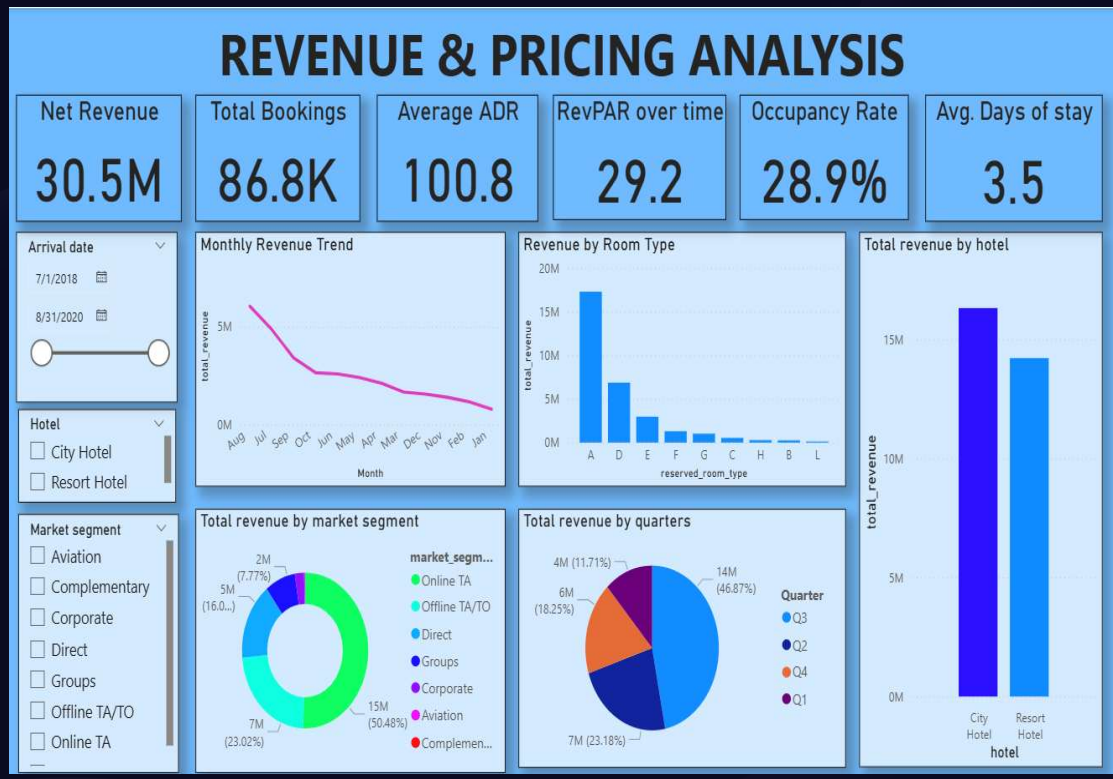
This analysis focuses on revenue performance and pricing strategy across the hotel portfolio.

## Revenue Distribution Analysis

- Revenue by hotel type (City vs Resort)
- Revenue by room type
- Revenue by market segment

## Key Insights

- City hotels generate higher revenue than resort hotels
- Certain room types contribute more revenue
- Online TA generates highest revenue (~50%)




**Conclusion:** This indicates that pricing and demand vary across segments and require optimization.

# Quarterly Revenue Trends

## Revenue Distribution Analysis

Revenue distribution across quarters demonstrates strong seasonal patterns in hotel demand.

### Quarterly Breakdown

- Q3 contributes highest revenue (~46%)
- Q1 contributes lowest revenue

### Key Insights

- Strong seasonal variation in hotel demand
- Peak seasons significantly impact revenue

**Conclusion:** Hotels should adopt dynamic pricing based on seasonal demand.

# Operational Insights

## Stay Pattern Analysis

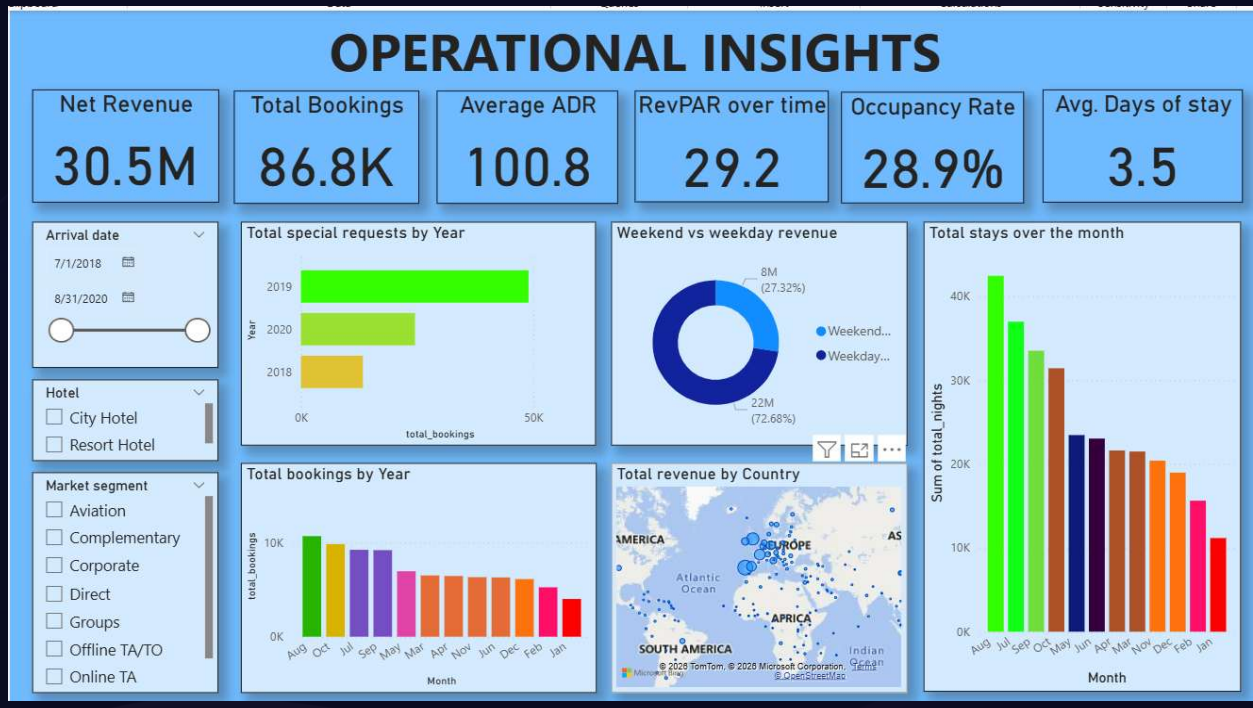
Operational analysis focuses on stay patterns and revenue distribution across the booking cycle.

### Revenue Distribution

- Weekday revenue (~73%)
- Weekend revenue (~27%)
- Average stay: 3.5 days

### Key Insights

- Business travelers dominate weekday bookings
- Weekend occupancy is comparatively low
- Short stay duration indicates quick turnover




Conclusion: This indicates that operational performance is heavily driven by weekday business demand, with significant opportunities to improve weekend occupancy and optimize stay duration.

# Key Insights

Key findings from the analysis:

### Low Occupancy Rate

Despite high revenue generation, occupancy rates remain below optimal levels

### Very Low Repeat Customers

Repeat customer rate is approximately ~3%, indicating poor customer retention

### High Dependency on Channels

Strong reliance on travel agents and Online TA for bookings

### City Hotels as Revenue Drivers

City hotels are the main revenue contributors across all segments

### Seasonal Trends

Strong seasonal trends significantly affect overall performance

### Weekday Dominance

Weekday bookings dominate over weekend occupancy patterns

**Highlight:** These insights reveal gaps in customer retention, pricing, and operational efficiency.

# Conclusion

This project demonstrates how data analytics helps convert raw hotel booking data into meaningful insights. The use of Power BI for visualization and decision-making enables stakeholders to identify trends, patterns, and opportunities for improvement across all operational dimensions.

## Key Takeaways

- Revenue performance analysis reveals significant variation across hotel types and market segments
- Customer booking behavior understanding shows heavy reliance on new customers and third-party channels
- Pricing strategy evaluation indicates need for dynamic pricing based on seasonal demand
- Operational insights identification highlights weekday dominance and low weekend occupancy

These insights support data-driven decision-making and help improve overall hotel performance.